

Notes To Myself on Arc Length

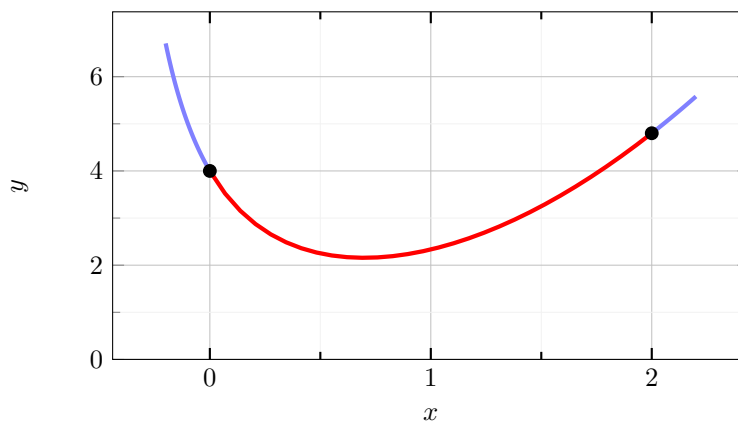
L.Dodd (v 1.3 on 2025/01/17 11:39:39)

1. Example

Consider function $f(x)$ (a plot is given below)

$$f(x) = x^2 + \frac{4}{2x+1}$$

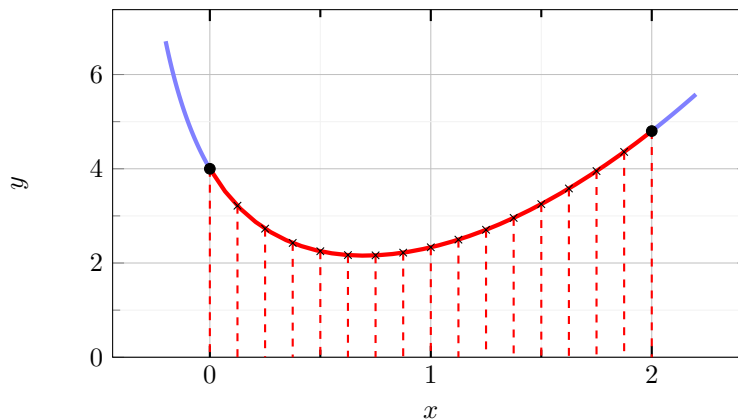
Say that we want the length along the function on the interval $[0, 2]$. That is, say want the length of red curve below



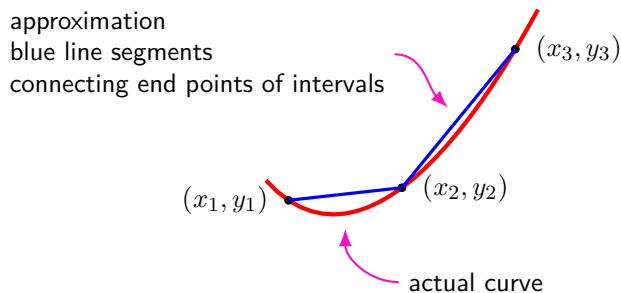
That curve is not a simple geometric figure. Finding a value for its length requires some additional work. We begin by partitioning $[0, 2]$ into n equally-sized intervals

$$0 = x_0 < x_1 < x_2 < \dots < x_n = 2$$

The graph below is for $n = 16$

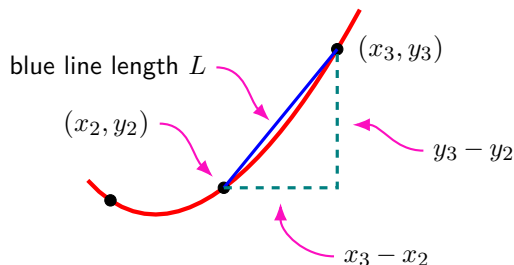


Within each interval we approximate the curve with a line segment. Here are is a close-up of an interval



The length of any line segment will be the hypotenuse of the right triangle connecting the end points of the interval. For instance, the length of the second line segment is

$$L = \sqrt{(y_3 - y_2)^2 + (x_3 - x_2)^2}$$



Thus, the length of the function between (x_2, y_2) and (x_3, y_3) is approximated by L . We repeat this for all pairs of points on the interval. Summing up the length of these line segments we get an approximation for the true arc length s

$$s \approx \sum_{i=1}^n \sqrt{(y_i - y_{i-1})^2 + (x_i - x_{i-1})^2}$$

Each term of the summation rewritten as follows

$$s \approx \sum_{i=1}^n \sqrt{1 + \left(\frac{y_i - y_{i-1}}{x_i - x_{i-1}} \right)^2} (x_i - x_{i-1})$$

or

$$s \approx \sum_{i=1}^n \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i} \right)^2} \Delta x_i$$

where

$$\Delta x_i \equiv x_i - x_{i-1} \quad \Delta y_i \equiv y_i - y_{i-1} \quad \text{for} \quad i = 1, n$$

One could use the above to approximate the arc length. All we need is $f(x)$ and a value for n . It would be quite tedious but the summation would converge.

An alternative formation is made possible with the Mean Value Theorem which states that there exists a $c_i \in [x_{i-1}, x_i]$ such that

$$\frac{\Delta y_i}{\Delta x_i} = f'(c_i) = \left. \frac{df}{dx} \right|_{x=c_i}$$

So the arc length *could* be written as

$$s \approx \sum_{i=1}^n \sqrt{1 + [f'(c_i)]^2} \Delta x_i$$

The advantage of the above formulation is that in the limit as the number of intervals n increases the the result above becomes this integral

$$s = \int_{x=a}^{x=b} \sqrt{1 + [f'(x)]^2} dx$$

In our case, $a = 0$ and $b = 2$ and

$$f(x) = x^2 + \frac{4}{2x+1} \quad \implies \quad f'(x) = 2x - \frac{8}{(2x+1)^2}$$

This means our arc length would be the value of this definite integral

$$s = \int_0^2 \sqrt{1 + \left[2x - \frac{8}{(2x+1)^2}\right]^2} dx$$

The above is very messy and (as far as the author knows) can NOT be done analytically. Numerical integration gives

$$s = 5.10510357182 \dots \approx 5.1$$

2. General

In general the arc length s for any continuous function $f(x)$ between points $x = a$ and $x = b$ is given by this integral

$$s \equiv \text{arc length (between } a \text{ and } b) = \int_a^b \sqrt{1 + \left(\frac{df}{dx}\right)^2} dx$$

Also, the *differential arc length* ds of $f(x)$ at x is given by

$$ds \equiv \text{differential arc length (at } x) = \sqrt{1 + \left(\frac{df}{dx}\right)^2} dx$$